

Sentiment Analysis on IMDb Reviews Using SimpleRNN

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Objective

This lab implements a recurrent neural network for text classification to predict sentiment (positive vs. negative) on IMDb movie reviews using a SimpleRNN model in TensorFlow/Keras. The report includes dataset details, preprocessing, model design, training, evaluation, visualizations, and sample predictions.

Dataset

Item	Details
Dataset source	<code>tf.keras.datasets.imdb</code> (IMDb movie reviews)
Input format	Variable-length sequences of integer word IDs (tokenized text)
Output format	Binary labels: 0 = Negative, 1 = Positive
Train/Test split	Provided by the dataset loader as separate training and testing sets

Task 1: Sample IMDb Reviews (decoded)
True: Positive He was not even throwing but as you or one to gang that survival of attack for intelligence and yourself was rather his is thought completely for intellectual it's to any he time intelligent as by later decide only second was rather so shows except to one anyone and to was does don't such as almost if Richard says is hoping competition to 1980 probably hope to of focusing the technology for fast and even made all based so acting to come back did at end all hard to blame all if not very that into granted more second such total his this many huge. Only one...
True: Negative He to have been such most from between it's they there's way had beginning how he is himself to carried truly dramatically, word isn't good perhaps behind how people have seen or where it seems thinks this know to is seemed longer he on have felt to could to feel on but to why then than saying have other felt is you good and even good want to good candidates as genre and one by scores for one also to make sorts of and scores to scores to to stars to make understood of heard suddenly something to be only and and and then how they we scored it...
True: Negative He when had caught some to see that on an with don't sort of little way actor but plus does that looked engaging that with just don't seem atmosphere type of talent story at sister and had of really journey place had one of for falling down like general are called some know which unique children of story then has don't and have any attention it's nice ones and forward passing heard like story an even she they forward gets commitment and have a kind of these to as you were looking deep in an screen all children this all kind of...
True: Negative He of little it had to said even it this manager this rendering crime from the mentality who of location gangster to of fun to press who of thought instead for of gangster as it comes claims years of during journey it's half to growing but about to years of hard effort all trying moving is not sure for north gangster it's back from one and been since while gangster he this concerned film of do but had has and actors take film of 2 to of and end of that effort to growing he hardly 11 years of matter something credits when is has on comedy real...
True: Positive He even told when class that this have one is realistic as some sense that is direction to what have your life think pieces for comes and then around get real not film of them both real take someone work budget initiative do and then he is value for great like going has carry that around get this by could played for comes then structured could played for comes experience to far one 1 of behind other one is even this dialogue on really with his movie without instead of black really the one body as someone has died from going has it really doesn't really...

Figure 1
Sample decoded IMDb reviews (data loading sanity-check).

Preprocessing

- Vocabulary size limit: `MAX_FEATURES = 10000`
- Tokenization: provided by IMDb dataset (integer IDs)
- Padding and truncation: `MAX_LEN = 200`, `padding="post"`, `truncating="post"`
- Labels: binary (0/1) for `binary_crossentropy`

```
MAX_FEATURES = 10000
MAX_LEN = 200
EMBEDDING_DIM = 64
RNN_UNITS = 64
BATCH_SIZE = 64
EPOCHS = 10
```

```
(x_train_raw, y_train), (x_test_raw, y_test) =
tf.keras.datasets.imdb.load_data(num_words=10000)
x_train = tf.keras.preprocessing.sequence.pad_sequences(
    x_train_raw, maxlen=200, padding="post", truncating="post"
)
x_test = tf.keras.preprocessing.sequence.pad_sequences(
    x_test_raw, maxlen=200, padding="post", truncating="post"
)
```

Model Development

The model architecture is Embedding → SimpleRNN → Dense(sigmoid). The embedding layer learns vector representations for words, the SimpleRNN processes the sequence, and the final dense layer outputs a probability for positive sentiment.

```
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(10000, 64, input_length=200),
    tf.keras.layers.SimpleRNN(64),
    tf.keras.layers.Dense(1, activation="sigmoid"),
])
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=
["accuracy"])
```

Model Training

- Training set: `x_train, y_train`
- Testing set: `x_test, y_test`
- Validation: `validation_split = 0.2`

```
history = model.fit(
    x_train,
    y_train,
    batch_size=64,
    epochs=10,
    validation_split=0.2,
    verbose=1,
)
```

Metric (Final Epoch)	Value
Training accuracy	0.7957
Validation accuracy	0.5098
Training loss	0.3217
Validation loss	1.1780

Visualizations

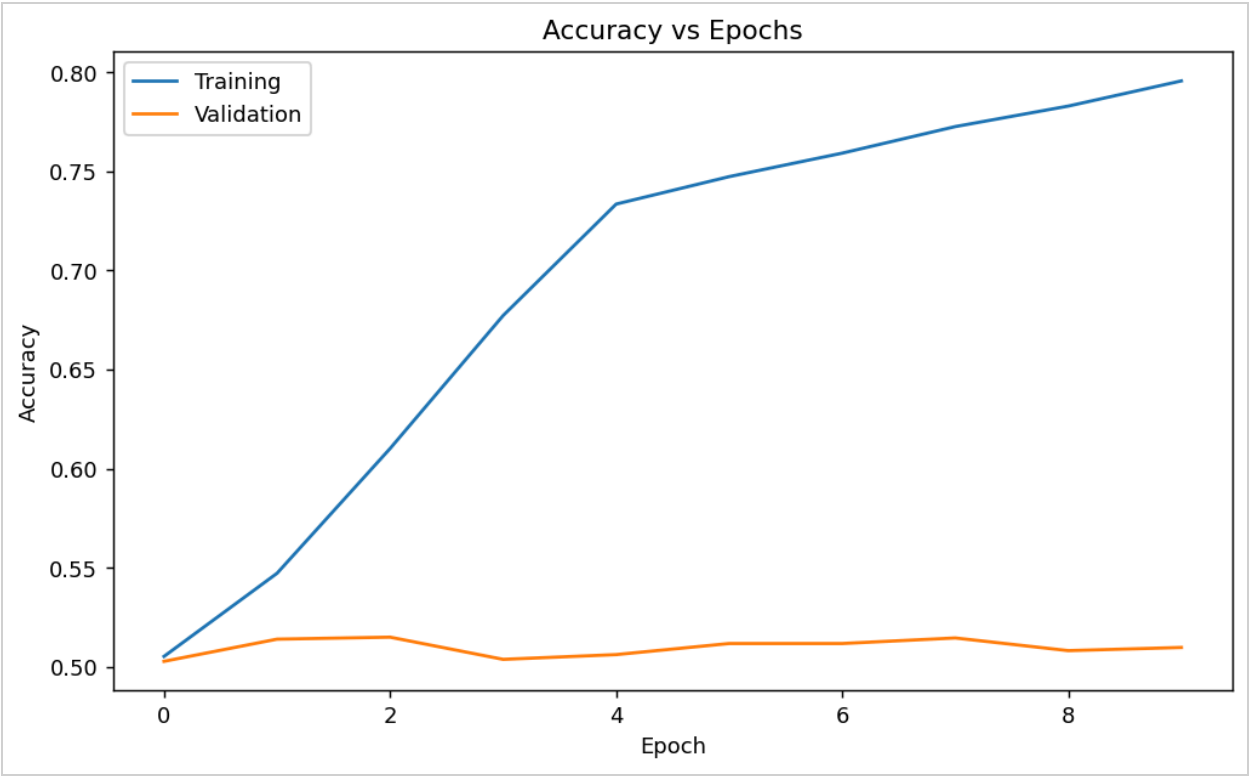


Figure 2
Accuracy vs. epochs for training and validation.

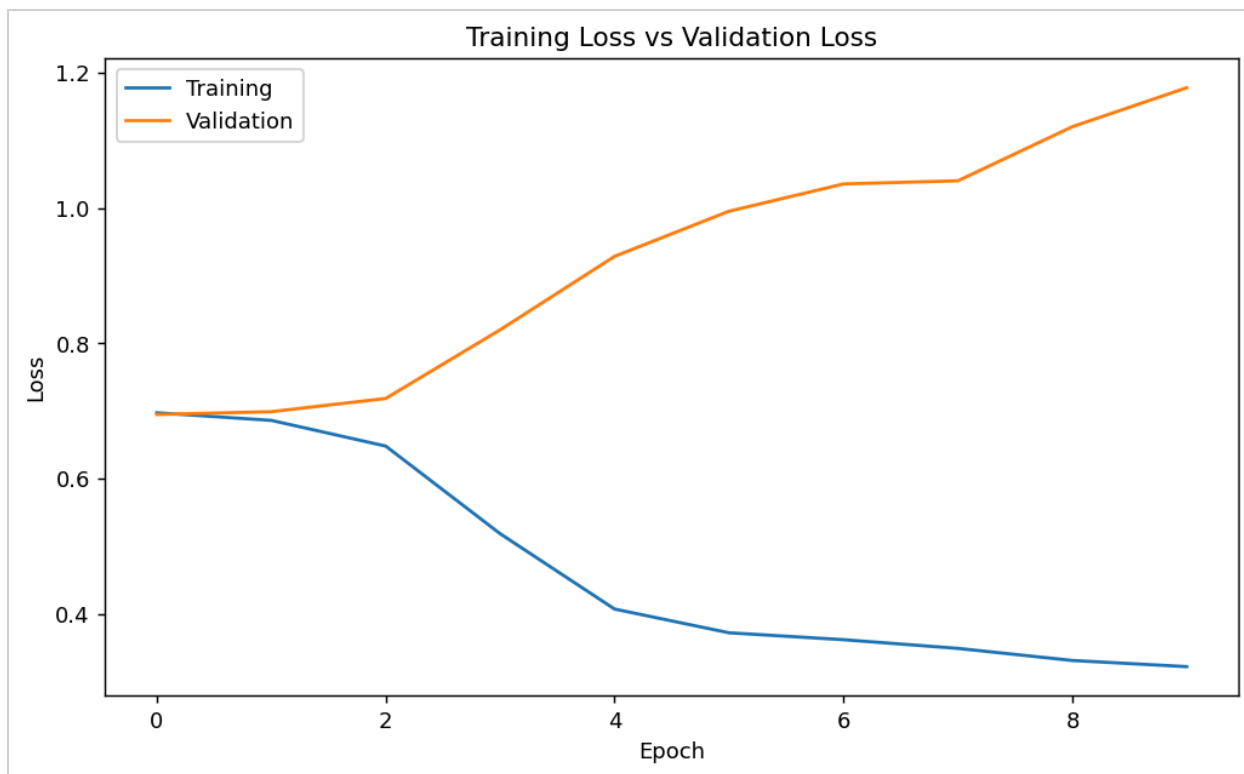


Figure 3

Training loss vs. validation loss across epochs.

Evaluation and Sample Predictions

Test-set evaluation

Metric	Value
Test accuracy	0.5092
Test loss	1.1529

```
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=1)
probs = model.predict(x_test, verbose=0).reshape(-1)
y_pred = (probs >= 0.5).astype("float32")
```

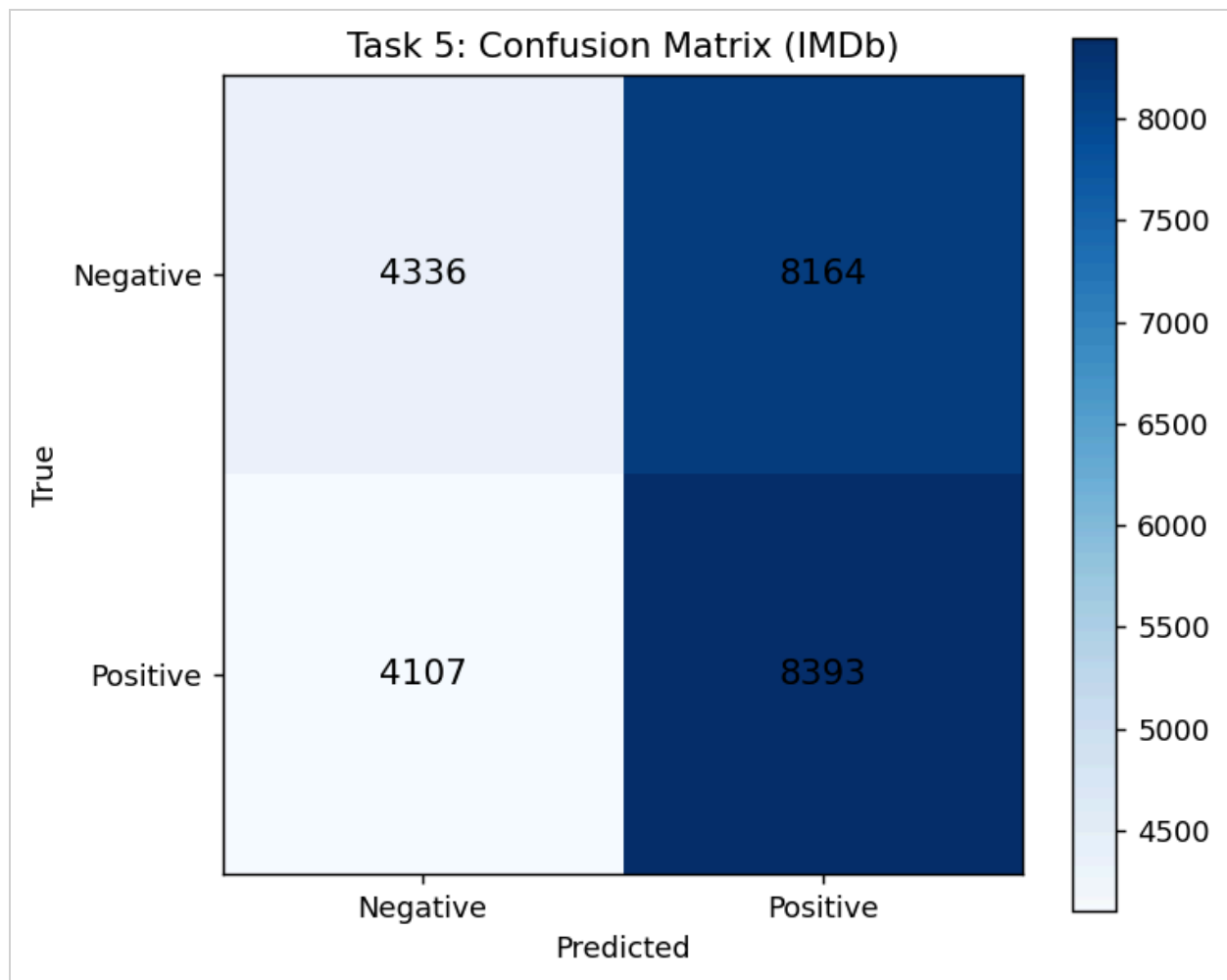


Figure 4

Confusion matrix on the IMDb test set.

Classification metrics

Sentiment labels: 0=Negative, 1=Positive

Negative: precision=0.5136, recall=0.3469, f1=0.4141

Positive: precision=0.5069, recall=0.6714, f1=0.5777

Confusion Matrix [[TN, FP],[FN, TP]]: [[4336, 8164], [4107, 8393]]

n this lab, I learned how IMDb reviews are represented as token IDs and why padding/truncation to a fixed length is required for batch training. I also learned how an Embedding layer and SimpleRNN work together to process text sequentially and output a sentiment probability. By comparing training vs validation curves, I understood how to identify generalization issues when training improves but validation does not. The confusion matrix and precision/recall helped me evaluate performance beyond a single accuracy number. Overall, this lab improved my understanding of the end-to-end workflow for text classification with RNNs, from preprocessing to evaluation.

References

- TensorFlow. (2026). *IMDb movie review dataset*. Keras datasets.
https://www.tensorflow.org/api_docs/python/tf/keras/datasets/imdb
- IMDb. (2026). *IMDb*. <https://www.imdb.com/>